SMART INDIA HACKATHON 2026 · PROBLEM STATEMENT SIH26261

VERSION 2.4 · TECHNICAL DOSSIER

VERITAS

AI-Powered Application Validation, 3-Layer Document Forensics & Infrastructure Compliance Verification Platform for AICTE

An explainable, evidence-linked pre-scrutiny copilot engineered for the All India Council for Technical Education (AICTE), Ministry of Education, Government of India. Veritas automates Approval Process Handbook (APH 2024–2027) validation, detects multi-layer PDF and image tampering, verifies certificates with issuing authorities, and hunts cross-college duplicate infrastructure photos while keeping final regulatory authority with human officials.

Organization & Domain

AICTE (Ministry of Education, Govt. of India)
Theme: **Smart Automation**
Category: **Software (AI / CV / Forensics)**

Target User Personas

1. **AICTE Scrutiny Officers & EVC**
2. **AICTE Competent Authorities**
3. **Institution Nodal Officers**

Repository & Stack

Next.js 15 + TypeScript + FastAPI
PyMuPDF · pikepdf · OpenCV · SIFT · imagehash · exifread · pgvector

1. Executive Product Vision

Every academic cycle, AICTE processes thousands of institutional dossiers for **Extension of Approval (EoA)**, **New Technical Institutions (LOA)**, and **Increase in Intake**. Each dossier comprises multi-page statutory certificates (State Government NOCs, Fire Safety Certificates, Sworn Affidavits, Architect-Certified Floor Plans) alongside geotagged photographs of laboratories, classrooms, and barrier-free facilities. Manual scrutiny of these voluminous submissions is slow, repetitive, and vulnerable to subtle document alterations or cross-college photo recycling.

Veritas transforms this workflow by acting as a rigorous, automated pre-scrutiny copilot. Before an AICTE Scrutiny Committee convenes, Veritas ingests the entire institutional dossier, extracts structured fields via layout-aware OCR, reconciles data across four independent sources, executes a 3-layer forensic inspection on every PDF, queries authorized issuing registries, and runs computer-vision & perceptual-hash checks on every facility photograph. Crucially, Veritas never issues black-box rejections: every AI finding is presented as an **evidence-linked advisory flag** tied to exact bounding-box coordinates, APH clause citations, and an immutable SHA-256 audit log.

2. Comprehensive Dossier Table of Contents (Pages 1–11)

Page	Section Title	Key Topics Covered
Page 1	Executive Overview & Product Vision	Problem statement metadata, executive abstract, core thesis, and complete dossier index.
Page 2	Problem Analysis & AICTE APH 2024–27 Norms	6 manual scrutiny bottlenecks, statutory APH space/equipment/faculty norms, and regulatory guardrails.
Page 3	Core USPs & Comparative Differentiation	6 technical USPs (QR-vs-OCR Cross-Lock, 3-Layer X-Ray, Clone Hunter, EXIF GPS) & benchmark comparison.
Page 4	System Architecture & Technology Stack	Multi-tier architecture diagram, open-source forensic research lineage, and full stack breakdown.
Page 5	End-to-End Process Flow & User Workflows	6-stage operational pipeline diagram, Nodal Officer self-scrutiny flow, and Scrutiny Officer workflow.
Page 6	Deep Dive: OCR Consistency & 3-Layer Forensics	4-way reconciliation logic, QR decoding, PDF incremental save/font forensics, ELA, Noise Grid & SIFT Copy-Move.
Page 7	Deep Dive: Authority Sync, CV & Clone Hunter	Issuing portal connectors, 64-bit pHash/dHash/wHash + 128-D pgvector search, and WGS84 Haversine GPS math.
Page 8	UI/UX Walkthrough (Part 1: Queue & Checklist)	Daylight Civic Portal design philosophy, Page 1 Scrutiny Queue, and Page 2 Application Checklist views.
Page 9	UI/UX Walkthrough (Part 2: Forensics & CV)	Page 3 Document Forensics Studio, Page 4 Infrastructure & Clone Comparator, and Page 5 Upload & Audit Portal.
Page 10 & 11	Database Schema, REST APIs, Built MVP vs. Phase 2 Roadmap	PostgreSQL + pgvector DDL, FastAPI REST endpoints, synthetic forensic dossier specs, and MVP vs. Phase 2 matrix.

AICTE Scrutiny Challenges & APH 2024–2027 Compliance Norms

Under the **AICTE Approval Process Handbook (APH) 2024–2027**, technical institutions applying for Extension of Approval (EoA), increase in intake, or establishment of a new institution must submit documentary and visual evidence demonstrating compliance across land, built-up carpet area, laboratory equipment, faculty-student ratios, fire safety, and barrier-free accessibility.

1. Six Critical Pain Points in Current Manual Scrutiny

1. Cross-Document Data Contradictions

An institution may apply for **180 seats** in its online application form while its State Government NOC only sanctions **120 seats**, or its architect floor plan only supports 43 students in Classroom CR-102. Catching these across 15+ PDFs per college is tedious and error-prone.

2. Post-Issuance PDF & Raster Tampering

Applicants occasionally alter dates or intake numbers on scanned or born-digital certificates using PDF editors or Photoshop—overwriting text spans, splicing signatures, or copying-and-pasting official rubber seals while leaving the original embedded QR code untouched.

3. Issuing Authority Verification Delays

Manually confirming whether a Municipal Fire Safety Certificate (`FIRE-MH-PUN-2023-4102`) or State NOC is genuinely active or expired in the issuing department's registry delays scrutiny committees and creates backlogs.

4. Cross-College Recycled Infrastructure Photos

Unscrupulous applicants may reuse high-end computer lab or workshop photographs submitted by another institution in a different city (slightly cropped, brightness-adjusted, or with a pasted college banner) to mask infrastructure shortfalls.

5. Manual Infrastructure & Floor Plan Checks

Verifying whether each classroom meets the mandatory **66.0 sq.m** carpet area norm (for 60 students), whether computer centers meet the **150.0 sq.m** and **1:6 PC ratio**, and whether wheelchair ramps meet the **1:12 slope** requires painstaking visual inspection.

6. Traceability & Auditability Gaps

When scrutiny observations are recorded across disconnected spreadsheets or notes, authorities lack an immutable, tamper-evident audit trail linking each deficiency notice or approval to the exact rule version, source response, and officer rationale.

2. Configurable AICTE Approval Process Handbook (APH 2024–2027) Rule Matrix

Veritas encodes statutory norms from the AICTE Approval Process Handbook into a configurable rule engine so every uploaded document, blueprint, and campus photograph is evaluated against exact quantitative thresholds:

APH 2024–27 Reference	Regulatory Domain	Prescribed Statutory Norm	Veritas Automated Verification Method
Chapter II · Clause 2.14	Sanctioned Intake & State NOC	Applied intake must not exceed State Govt NOC sanctioned intake; valid for current AY.	4-Way OCR vs. QR Payload vs. State e-NOC API vs. AICTE Historical EoA DB.
Appendix-4 · Clause 4.2	UG Lecture Classroom Space	Minimum **66.0 sq.m** carpet area per division of 60 students (~**1.1 sq.m/student**).	Architect Floor Plan OCR schedule parser + CV classroom desk/seating counter.
Appendix-4 & Appendix-5	Central Computer Center & PCs	Minimum **150.0 sq.m** carpet area; PC-to-student ratio of **1:6** (min 20–30 PCs per lab).	Floor plan dimension check + CV workstation monitor detection + pHash Clone Hunter.
Annexure-3 · Clause 4.1	Fire & Life Safety Compliance	Active Fire Safety NOC from Municipal/State Fire Authority; wet riser, extinguishers & sprinklers.	Fire Portal API expiry sync + SIFT seal copy-move check + CV fire gear detection.
Chapter VII · Clause 7.9	Barrier-Free Environment	Wheelchair ramp with max **1:12 gradient**, dual handrails, and tactile floor guidance tiles.	CV ramp geometry, handrail & tactile paving detection + EXIF GPS on-campus check.
Chapter VII · Clause 7.1	Faculty-Student Ratio (FSR)	Minimum **1:20** regular faculty-to-student ratio for UG Engineering programmes.	Cross-check declared faculty list against AICTE PAN/APAAR Registry & applied intake.
Annexure-2 · Undertaking	Sworn e-Stamp Affidavit & TER	Sworn affidavit on **INR 100/- non-judicial e-Stamp** with valid Class-III DSC & TER fee paid.	StockHolding CRA e-Stamp check + PKCS#7 `/Sig` verification + Bharatkosh TER receipt match.

Statutory Safeguard Mandate (Advisory Decision Support Only)

In strict compliance with SIH26261 guidelines, Veritas treats all AI/CV/Forensic outputs as **preliminary, evidence-linked advisory indicators**. Veritas never auto-rejects an institution; instead, it equips AICTE Scrutiny Officers and Expert Visit Committees (EVC) with visual proof so human authorities can confirm deficiencies, order physical inspections, or override flags with logged justification.

Six Core USPs & Competitive Differentiation Matrix

Existing document portals rely either on manual visual inspection or on generic OCR-to-LLM wrappers that blindly trust whatever text is printed on a page. If an applicant edits a PDF in Photoshop to change `120 Seats` to `180 Seats`, a standard OCR/LLM pipeline simply extracts `180 Seats` and marks the application compliant. **Veritas is engineered specifically to defeat such failure modes through six orthogonal USPs:**

USP 1 · Cryptographic QR-vs-OCR Cross-Lock

Anti-Forgery

Government certificates (e-NOCs, e-Stamps, Municipal Fire NOCs, Architect Blueprints) embed QR codes containing signed JSON or URI payloads. Veritas decodes the QR payload via multi-pass OpenCV thresholding and cross-locks every attribute (`approved_intake`, `valid_upto`, `cert_no`) against the printed OCR text on the document face, drawing a direct visual connector between the QR code and any altered text span.

USP 2 · 3-Layer "X-Ray" Document Forensics

Pixel + Container

Combines container-level PDF inspection (`pikepdf` incremental `%%EOF` trailer counts, `/Producer` software checks, `/CreationDate` vs `/ModDate` deltas, and `PyMuPDF` span-level font anomaly detection) with pixel-level **Error Level Analysis (ELA)**, **8×8 Laplacian Noise Variance Grids**, and **SIFT Keypoint Copy-Move Detection** to catch pasted signatures and cloned seals.

USP 3 · Cross-College Clone Hunter (`pHash` + `pgvector`)

National Scale

Computes 64-bit Perceptual DCT Hashes (`pHash`), Difference Hashes (`dHash`), Wavelet Hashes (`wHash`), and 128-dimensional L2-normalized spatial color/gradient embeddings for every uploaded campus photo. Searches across all historical and multi-institution submissions in `pgvector` to catch recycled, cropped, or brightness-adjusted photos from other colleges.

USP 4 · EXIF Geo-Temporal Boundary Verifier

Geospatial

Extracts raw EXIF GPS coordinates (`GPSLatitude`, `GPSLongitude`), capture timestamps, and editing software tags from uploaded facility photos. Computes the exact WGS84 Haversine distance in meters against the institution's registered AICTE campus coordinates (e.g., flagging a photo taken 624.1 km away in Nagpur for a Pune college).

USP 5 · Dual-Role Pre-Submission Self-Scrutiny

Proactive

Empowers **Institution Nodal Officers** to run the exact same validation & forensic pipeline in a self-check sandbox *before* formal submission to AICTE, drastically reducing accidental omissions, expired uploads, and avoidable deficiency show-cause cycles.

USP 6 · SHA-256 Hash-Chained Regulatory Ledger

Audit-Ready

Every rule evaluated, AI confidence score, issuing authority API response, and Scrutiny Officer override is appended to an immutable cryptographic hash chain (`prev_hash` → `entry_hash`), providing RTI- and court-ready regulatory transparency.

2. Competitive & Functional Comparison Matrix

Capability / Verification Vector	Legacy Manual Scrutiny	Generic OCR / LLM Wrapper	Veritas (AICTE SIH26261 Platform)
Multi-Document Field Reconciliation	Manual side-by-side reading (30–45 mins/dossier)	Extracts text, but lacks historical AICTE DB & QR cross-lock	**Automated 4-Way Matrix** (App Form ↔ OCR ↔ QR/API ↔ AICTE DB in <3 sec)
Altered Text on Authentic Certificate	Frequently missed if visually aligned	Fails completely (reads altered text as truth)	**Caught via 3 Vectors:** QR-vs-OCR mismatch + Span Font Anomaly + ELA Hotspot
Cloned Official Stamps / Signatures	Undetectable to naked eye	Not supported	**Caught via SIFT Keypoint Copy-Move** + 8×8 Laplacian Noise Variance Grid
Recycled Lab Photos from Other Colleges	Impossible to remember across 8,500+ colleges	Exact MD5 file hash only (defeated by 1% crop)	**Caught via 64-bit pHash/dHash/wHash + 128-D pgvector** (survives crop & brightness edits)
Off-Campus / Old Facility Photographs	Not verified systematically	Not supported	**Automated EXIF GPS Haversine Distance Check** against 500m campus boundary
Issuing Authority Source Verification	Manual phone/email/portal lookup	Not integrated	**Live Gateway Connectors** with 5 statutory statuses & latency telemetry
Explainability & Human Governance	Subjective notes	Opaque "Pass/Fail" prompt output	**Bounding-Box Evidence Links + APH Clause Citations + SHA-256 Audit Log**

Multi-Tier Architecture, Open-Source Lineage & Technology Stack

Veritas follows a modular, decoupled microservices-ready architecture pairing a **Next.js 15 (App Router) TypeScript Civic Portal** on the frontend with a high-throughput **Python FastAPI AI, Forensics & Computer Vision Engine** backed by **PostgreSQL (`pgvector`) / SQLite** storage.

FIGURE 4.1 · VERITAS 5-TIER SYSTEM ARCHITECTURE



1. Complete Technology Stack Specification

Layer	Technologies & Libraries	Role & Engineering Justification in Veritas
Frontend Portal	`Next.js 15.1` (App Router), `React 19`, `TypeScript 5.7`, `Tailwind CSS 3.4`, `Lucide Icons`	Server-rendered and client-interactive 5-page Civic GovTech portal with responsive SVG bounding-box overlays, live ELA parameter controls, and role-based workflows.
Backend API	`Python 3.12+`, `FastAPI`, `Uvicorn`, `Pydantic v2`, `python-multipart`	High-concurrency asynchronous REST backend serving dossier validation reports, live ELA recomputation, authority API pings, and multipart file upload diagnostics.
PDF & OCR Forensics	`PyMuPDF (fitz) 2.0`, `pikepdf 10.17` (QPDF C++ bindings), `pyHanko` patterns	High-DPI PDF page rasterization, span-level font/coordinate extraction (`get_text("dict")`), `/Producer` & `/ModDate` inspection, `%%EOF` incremental save counting, and PKCS#7 `/Sig` checks.
Image Forensics & QR	`OpenCV 5.0` (`cv2`), `NumPy`, `SciPy`, `Pillow 13.0`, `qrcode`	Multi-pass `QRCodeDetector`, JPEG recompression Error Level Analysis (ELA), 16×16 block z-score clustering, 8×8 Laplacian noise variance grids, and `SIFT_create()` copy-move detection.
Clone Hunter & Geo-CV	`imagehash 4.3`, `PyWavelets`, `exifread 3.5`, `piexif`	Computes 64-bit `pHash`, `dHash`, `aHash`, and `wHash` fingerprints plus 128-D spatial color/Sobel vectors; extracts EXIF GPS tags for WGS84 Haversine campus boundary verification.
Database & Vector Store	`PostgreSQL 16` + `pgvector` (`vector(128)`), `SQLite3`, `hashlib (SHA-256)`	Production `schema_pgvector.sql` with IVFFlat cosine index (`vector_cosine_ops`) paired with zero-config local SQLite persistence and SHA-256 hash-chained audit logging.

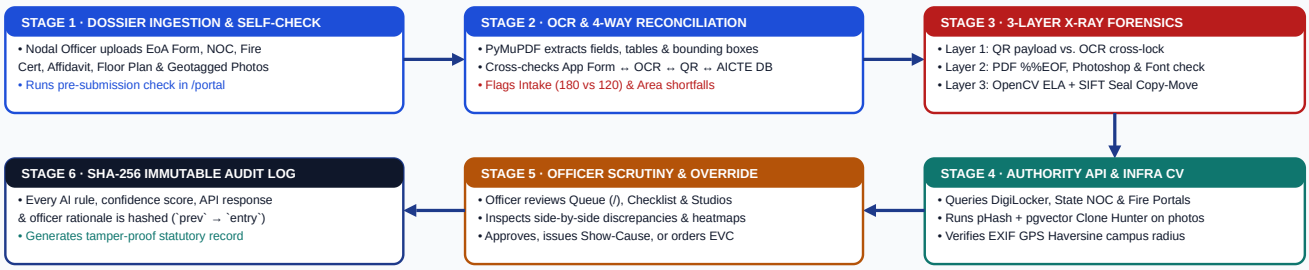
2. Open-Source Research Lineage Incorporated

Our forensic pipeline synthesizes peer-reviewed techniques from leading open-source repositories: **GuidoBartoli/sherloq** and **CodeRafay/Forensic-Image-Analysis-Toolkit** (multi-layer ELA, noise separation, and SIFT + RANSAC copy-move localization where findings are reported as explainable measurements rather than black-box verdicts), **mytechnotalent/pixelproof** (grid-based noise profiling and Adobe 8BIM/metadata inspection), and **pikepdf / pyHanko** (incremental PDF update and signature byte-range analysis).

End-to-End Application Scrutiny Process & Role Workflows

Veritas orchestrates the regulatory lifecycle across two primary user personas—the **Institution Nodal Officer** (pre-submission self-scrutiny and rectification) and the **AICTE Scrutiny Officer / Competent Authority** (official validation, forensic review, EVC routing, and final sign-off).

FIGURE 5.1 · SIX-STAGE END-TO-END VERIFICATION & SCRUTINY PIPELINE



1. Step-by-Step Operational Workflow for AICTE Staff

1. **Daily Triage on the Scrutiny Queue (`/`):** When an AICTE Scrutiny Officer logs in, they land on the **Scrutiny Queue**. Applications are ranked by their composite pre-scrutiny risk index and grouped into *All Dossiers*, *Needs Review*, and *Ready to Approve*. A Priority Findings sidebar surfaces the top critical anomalies across the officer's queue.
2. **Dossier & APH Checklist Review (`/applications/[appId]`):** Clicking an institution opens its full 4-way **Cross-Document Consistency Table** and **APH 2024–27 Checklist**. Here the officer sees at a glance where the application form conflicts with uploaded certificates or historical AICTE records, and can trigger live re-verification against Issuing Authority portals.
3. **Deep-Dive Certificate Verification (`/forensics`):** If a certificate is flagged for QR-vs-OCR mismatch, font injection, or seal cloning, clicking *Inspect Evidence* opens that exact document in the **3-Layer Document Forensics Studio**. The officer toggles between the Annotated View, QR Cross-Lock View, ELA Heatmap (with live JPEG quality & contrast sliders), and Seal Duplication View.
4. **Infrastructure & Clone Inspection (`/infrastructure`):** For facility flags, the officer opens the **Infrastructure & Clone Hunter** page to inspect side-by-side duplicate photo matches across colleges, computer-vision desk/PC counts, and EXIF GPS campus boundary distances.
5. **Human-in-the-Loop Disposition & SHA-256 Sign-Off:** After reviewing the visual evidence, the officer enters mandatory statutory remarks and commits one of three dispositions: (a) *Approve / Clear Advisory Flags*, (b) *Order EVC Physical Inspection*, or (c) *Issue Deficiency / Show-Cause Notice*.

2. Role-Based Access Control (RBAC) & Safeguard Matrix

User Role	Primary Portal Views	Permitted Actions	Statutory Safeguard
Institution Nodal Officer	`/portal` (Pre-Submission Upload Lab), Own Institution Checklist	Upload draft PDFs & campus photos; view preliminary deficiency flags; submit rectification notes.	Cannot view other colleges' dossiers or alter official scrutiny statuses.
AICTE Scrutiny Officer	`/` (Queue), `/applications/[id]`, `/forensics`, `/infrastructure`	Inspect all forensic & CV layers; ping authority APIs; override AI flags; recommend EVC or Deficiency Notice.	Mandatory written rationale required for every override or adverse flag.
Competent Authority / EC	All Views + `/portal` Immutable SHA-256 Audit Ledger	Final statutory approval (EoA/LOA issuance), EVC directive confirmation, and full audit chain verification.	Cryptographic hash chain prevents retroactive alteration of audit logs.

Module 1 (OCR Consistency) & Module 2 (3-Layer X-Ray Document Forensics)

1. Module 1: Intelligent OCR & 4-Way Cross-Document Consistency Engine

Implemented in `backend/forensics_engine.py` and `backend/seed_generator.py`, Module 1 ingests uploaded PDF certificates and extracts both structured key-value fields and span-level geometric metadata using `PyMuPDF` (`page.get_text("dict")`). Every extracted text span is normalized to relative page coordinates `[x / W, y / H, w / W, h / H] ∈ [0, 1]`[4] and reconciled across four independent truth sources:

```
Reconciliation Tuple: R(field) = ( V_AppForm , V_PrintedOCR , V_QR_or_AuthorityAPI , V_HistoricalAICTEDB ) Flag Condition: Mismatch(field) = TRUE iff ∃ i, j ∈ {1..4} such that Normalize(V_i) ≠ Normalize(V_j)
```

- **Example Caught in Benchmark Dossier `AICTE-2026-EOA-8841`:** The online Application Form requests `180 Seats` for *B.Tech CSE (AI & ML)*, and the printed OCR text on Row 6 of `state_noc_8841.pdf` also reads `180 Seats (Annual Intake)`. However, the decoded QR payload on the same certificate and the Maharashtra DHTE API both report `approved_intake: 120`, matching AICTE's 2025–26 historical EoA record (`120 Seats`). Veritas immediately flags this 4-way contradiction.

2. Module 2: 3-Layer "X-Ray" Document Forensics Engine

Veritas inspects every uploaded certificate across three orthogonal forensic layers so that even sophisticated forgeries are exposed with concrete visual evidence:

Layer 1 · Multi-Pass QR-vs-OCR Cross-Lock (`decode_qr_codes` & `run_full_document_xray`)

PDF pages are rasterized at 150–300 DPI into RGB NumPy arrays. To handle low-contrast scans, stamps overlapping QR quiet zones, or uneven binarization, Veritas runs OpenCV's `cv2.QRCodeDetector().detectAndDecodeMulti()` across three preprocessing passes: (1) Raw RGB, (2) Luminance Grayscale, and (3) Otsu's Adaptive Bimodal Thresholding (`cv2.THRESH_BINARY + cv2.THRESH_OTSU`).

Once decoded, the JSON payload is parsed and compared field-by-field against all OCR spans on the page. When a discrepancy is found (e.g., QR `valid_upto: "2025-03-31"` vs. printed OCR `"2026-12-31"` on `DOC-8841-FIRE`), Veritas records both bounding boxes and renders an orthogonal SVG tether linking the QR code directly to the tampered text line.

Layer 2 · PDF Structural, Incremental Update, Font Span & DSC Inspection (`inspect_pdf_forensics`)

Using `pikepdf` (QPDF) and `PyMuPDF`, Veritas inspects the internal PDF container structure without relying on visual appearance:

- **Incremental Save Detection (`%%EOF` Counting):** Scans raw PDF byte streams for multiple `%%EOF` markers and XRef updates, proving a document was re-saved after initial generation.
- **Metadata & Producer Screening:** Compares `/CreationDate` against `/ModDate` and screens `/Producer` and `/Creator` against editing tools (`Adobe Photoshop`, `Illustrator`, `GIMP`, `Canva`, `iLovePDF`, `Foxit PhantomPDF`).
- **Span-Level Font Anomaly Detection:** Computes the character-weighted frequency distribution of fonts across the page to identify the dominant body font (e.g., `Helvetica`). Any numerical or date span rendered in an outlier font family (e.g., `180 Seats` injected in `Courier` at `12.5pt`) is automatically flagged with its exact bounding box.
- **Digital Signature (DSC) Check:** Inspects `/Type /Sig` and `/SubFilter /adbe.pkcs7.detached` byte-range structures.

Layer 3 · Visual Tamper Heatmap: ELA + 8×8 Laplacian Noise Grid + SIFT Copy-Move

```
ELA Pixel Energy: E(x,y) = | I_orig(x,y) - JPEG_Q(I_orig)(x,y) | * Scale Block Z-Score: Z(b_r,c) = ( Mean(E in 16x16 block) - μ_page ) / σ_page [Flagged if Z > 3.1 and Mean > 5.0] SIFT Copy-Move: Match(kp_i, kp_j) valid iff ||desc_i - desc_j||_2 < 0.78 * d_second AND ||pt_i - pt_j||_2 > d_min
```

- **Error Level Analysis (ELA):** Re-encodes the document at configurable JPEG quality *Q* (default 90%, adjustable live from 50% to 98%), computes absolute pixel differences, applies an Inferno forensic colormap, and morphologically clusters 16×16 blocks with *Z > 3.1* into tamper bounding boxes.
- **8×8 Laplacian Noise Variance Grid:** Convolves the grayscale page with a 3×3 Laplacian kernel ($\nabla^2 I$) to isolate high-frequency scanner/sensor noise, flagging grid cells whose noise standard deviation deviates >3.2× MAD from the page median.
- **SIFT Keypoint Copy-Move Forgery Detection:** Extracts up to 1,200 Scale-Invariant Feature Transform (`cv2.SIFT_create()`) keypoints, matches descriptors within the same image via `BFMatcher`, filters via Lowe's ratio test and minimum spatial separation, and clusters quantized displacement vectors (Δx, Δy) to expose duplicated official stamps (such as the cloned Chief Fire Officer seal on `DOC-8841-FIRE`).

Module 3 (Authority Source Sync) & Module 4 (Infrastructure CV & Clone Hunter)

1. Module 3: Issuing Authority Source Verification Engine

Forensic inspection of a PDF is necessary but not sufficient—a document can be freshly generated from scratch with clean metadata if an applicant fabricates the entire certificate. Module 3 closes this gap by querying authoritative external registries using the certificate's unique identifier (`cert_no`) and institution ID (`inst_id`), classifying every certificate into one of five statutory states:

Statutory Status	Definition & Trigger Condition	Scrutiny Routing Action
Verified	Issuing portal returns HTTP 200 and all attributes (`intake`, `valid_upto`, `inst_id`) match the uploaded document 100%.	Auto-marked compliant in APH checklist.
Mismatch Detected	Certificate exists in registry, but one or more attributes on the uploaded PDF differ from the official source record (e.g., `DOC-8841-NOC`).	Flagged critical; routed to Scrutiny Officer for Show-Cause Notice.
Expired	Registry confirms the certificate's validity date has lapsed prior to the current academic cycle (e.g., `DOC-8841-FIRE` expired `2025-03-31`).	Flagged critical; renewal certificate required.
Pending	Application or renewal is under processing at the issuing authority (e.g., Senate affiliation counter-signature awaited in `AICTE-2026-NEW-9014`).	Placed in conditional monitoring queue.
Manual Review Required	Issuing authority portal is offline, legacy non-digitized certificate, or inconclusive response.	Routed for manual verification / EVC physical inspection.

2. Module 4: Infrastructure Computer Vision & "Cross-College Clone Hunter"

Implemented in `backend/cv_engine.py`, Module 4 inspects every uploaded facility photograph and architectural floor plan across three sub-engines:

Sub-Engine 4A · Multi-Hash & 128-D Vector "Cross-College Clone Hunter"

Cryptographic hashes (MD5/SHA-256) fail the moment an image is re-saved at a different JPEG quality, slightly cropped, or brightened. Veritas implements a dual-stage perceptual & vector similarity search across all historical and multi-institution submissions:

```
1. Multi-Hash Fingerprints: pHash (64-bit 2D-DCT), dHash (Horizontal Gradient), aHash (Mean), wHash (Haar Wavelet) 2. Hamming Similarity: Sim_Hash = (1 - min(D_H(pHash), D_H(dHash), D_H(wHash)) / 64) * 100% [Match if D_H ≤ 12] 3. 128-D Spatial Embedding: 4x4 Grid × (3 RGB Means + 3 RGB Stds + 2 Sobel Gradient Moments) → L2-Normalized Vector v ∈ R^128 4. Cosine Similarity: Sim_Cos(u, v) = (u · v) / (||u||_2 ||v||_2) [Match if Sim_Cos ≥ 0.94]
```

Real Benchmark Catch: When *Vishwakarma Institute, Pune* uploads `computer_lab_8841.jpg` (which has been brightened by 3% and has a fake college banner spliced onto the top-left wall), Veritas computes `pHash = beb4c1c9c666341d` and matches it against `MEDIA-HIST-3109-LAB` submitted by *Deccan College of Engineering, Nagpur* (`pHash` Hamming distance = 8 bits, 128-D Cosine Similarity = 100.0%).

Sub-Engine 4B · EXIF WGS84 Haversine Geo-Temporal Boundary Verifier

Using `exifread`, Veritas extracts raw rational DMS tags (`GPSLatitude`, `GPSLongitude`), converts them to signed WGS84 decimal degrees, and computes the great-circle Haversine distance d from the institution's registered campus center (φ_{campus}, λ_{campus}):

```
a = sin²(Δφ / 2) + cos(φ_photo) · cos(φ_campus) · sin²(Δλ / 2) d = 2 · R_earth · atan2( √a , √(1 - a) ) where R_earth = 6,371,000 meters [Verified iff d ≤ 500.0m]
```

In `AICTE-2026-EOA-8841`, Classroom `CR-102` (d = 54.6m) and the Barrier-Free Ramp (d = 15.3m) pass within the 500m Pune campus boundary, whereas the cloned Computer Lab photo carries Nagpur GPS coordinates (21.1458°N, 79.0882°E)—**624.1 km away** from the Pune campus!

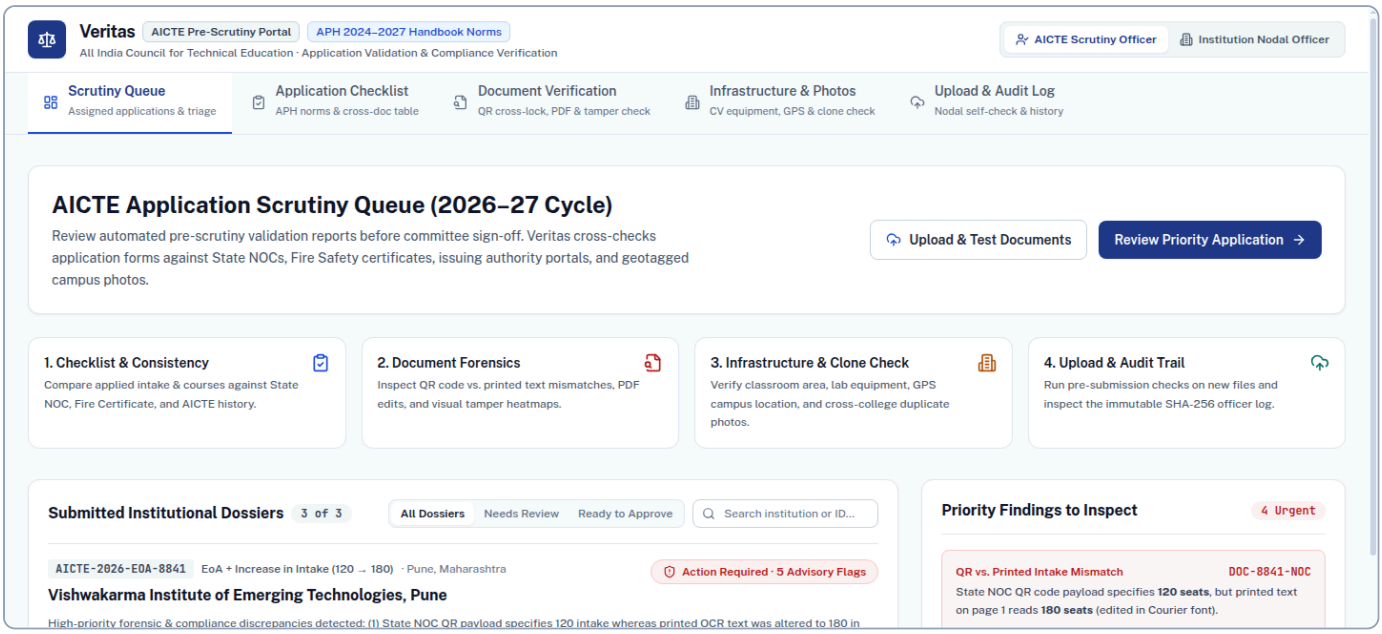
Sub-Engine 4C · AICTE APH Equipment & Carpet Area Compliance

Evaluates detected equipment counts and blueprint room schedules against APH 2024–27 norms: flagging `CR-102` for having only **38 student desks** and **48.0 sq.m** carpet area against the required **60 seats / 66.0 sq.m** division norm, and flagging the Computer Lab for showing **16 workstation terminals** against the required **30 terminals** (1:6 ratio).

Daylight Civic UX Architecture: Scrutiny Queue & Application Checklist

A core design principle of Veritas is **cognitive clarity for government officials**. Rather than overwhelming AICTE staff with a dark, single-screen terminal, Veritas uses a **daylight-first Civic GovTech design system** (`Public Sans` + `JetBrains Mono`, `#F8FAFC` slate-alabaster canvas, `#FFFFFF` cards, and `#1E3A8A` AICTE Navy accents) structured across five purpose-built pages.

1. Page 1 (`/`) — AICTE Application Scrutiny Queue (Landing Triage View)



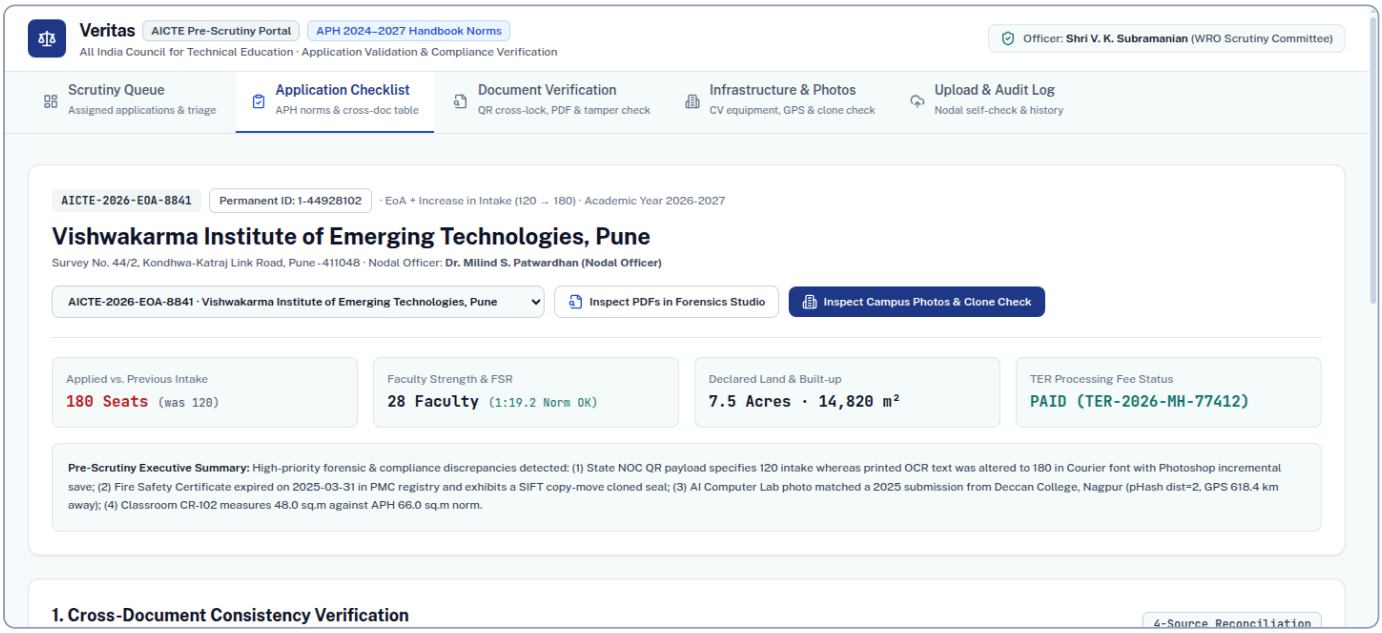
What AICTE Staff Sees First

Officers land directly on their **Assigned Applications Queue**. Filter tabs (*All Dossiers*, *Needs Review*, *Ready to Approve*) and instant search let staff prioritize high-risk applications (`AICTE-2026-EOA-8841`) while a 4-step guided banner explains the portal workflow.

Priority Findings & Gateway Health

The right-hand column surfaces urgent cross-document and forensic alerts (e.g., *QR vs. Printed Intake Mismatch*, *Expired Fire NOC*, *Cross-College Cloned Lab Photo*) with one-click deep links to the exact verification studio.

2. Page 2 (`/applications/[appId]`) — APH Checklist & 4-Way Consistency Workspace



4-Source Consistency Matrix & APH Checklist

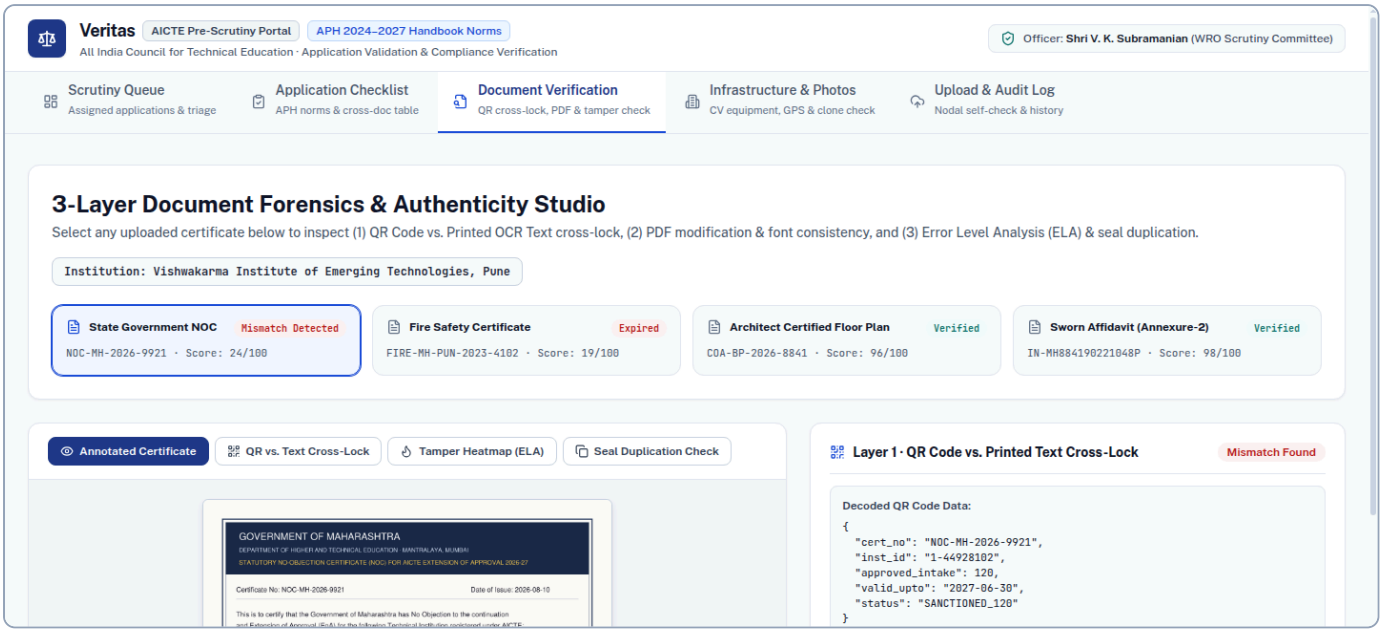
Presents the institution's key metrics (Applied vs. Previous Intake, FSR, Land/Built-up Area, TER Fee status), followed by the **4-Way Cross-Document Consistency Table** and the clause-by-clause **APH 2024–27 Checklist**.

Live Authority Sync & Officer Disposition

Officers can re-verify any certificate against its Issuing Authority portal in one click and record their official committee decision (*Approve*, *Order EVC Physical Inspection*, or *Issue Deficiency Notice*) at the bottom of the dossier.

Document Forensics Studio, Infrastructure CV & Nodal Self-Check Portal

1. Page 3 (`/forensics`) — 3-Layer Document Verification Studio



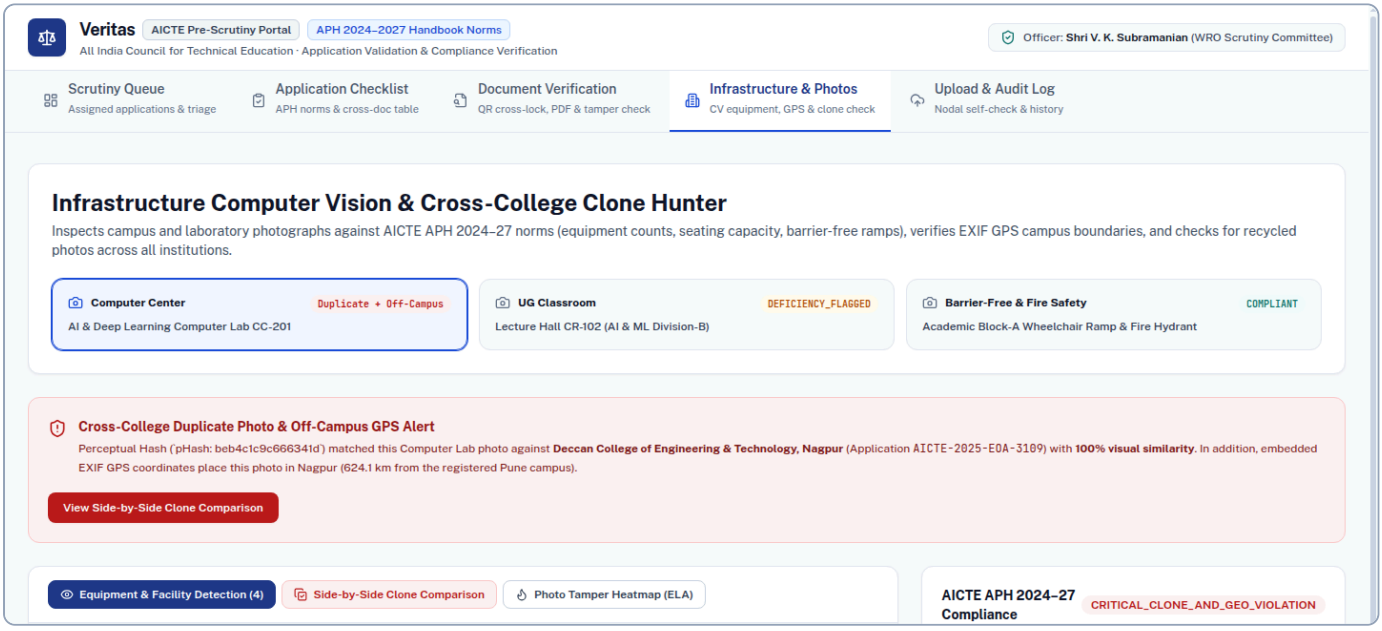
Interactive Certificate & Heatmap Viewer

Officers select any of the 4 uploaded PDF certificates at the top and toggle between **Annotated Certificate**, **QR vs. Text Cross-Lock** (with red SVG connector lines linking the QR code to altered text), **Tamper Heatmap (ELA)** (with live JPEG quality & contrast sliders), and **Seal Duplication Check**.

Plain-English 3-Layer Verification Report

The right column translates complex forensic signals into clear regulatory cards: **Layer 1** (Decoded QR JSON vs. Printed OCR Table), **Layer 2** (PDF editing software, incremental saves, font mismatches & DSC status), and **Layer 3** (ELA & SIFT copy-move summary).

2. Page 4 (`/infrastructure`) — Infrastructure CV & Cross-College Clone Hunter



Side-by-Side Cross-College Clone Comparator

When a duplicate photo is flagged (`MEDIA-8841-LAB`), officers can switch to the **Side-by-Side Clone Comparison** view to inspect the submitted Pune photo alongside the matched 2025 archive photo from *Deccan College, Nagpur* (`96.9% match`), clearly highlighting the pasted college banner.

Page 5 (`/portal`) — Upload Lab & SHA-256 Audit Log

The fifth page (`/portal`) provides a drag-and-drop **Live Pre-Submission Uploader** (plus 4 one-click benchmark test files) for Nodal Officers, paired with the complete **SHA-256 Hash-Chained Regulatory Audit Trail**.

PostgreSQL + pgvector Schema, REST API Catalog & Synthetic Test Suite

1. Relational & Vector Database Schema (`backend/schema_pgvector.sql` & `database.py`)

Veritas ships with both a production **PostgreSQL + `pgvector`** schema (`schema_pgvector.sql`) and a zero-configuration local **SQLite** persistence layer (`veritas.db`) containing five normalized tables:

Table Name	Primary Key & Indexes	Key Columns & Purpose
`institutions`	`inst_id` (PK)	`name`, `state`, `city`, `campus_lat`, `campus_lng`, `campus_radius_m` (500m default), `historical_eoa_intake`, `faculty_count`, `ter_payment_status`, `ter_receipt_no`, `nodal_officer`.
`applications`	`app_id` (PK), FK `inst_id`	`academic_year`, `category`, `applied_intake`, `applied_courses_json`, `declared_land_acres`, `declared_builtup_sqm`, `overall_status`, `risk_score`, `checklist_json`, `consistency_flags_json`.
`documents`	`doc_id` (PK), FK `app_id`	`doc_type`, `file_path`, `issuing_authority`, `cert_number`, `issue_date`, `expiry_date`, `authority_status`, `forensic_score`, `extracted_fields_json`, `qr_payload_json`, `pdf_metadata_json`, `forensic_findings_json`.
`infrastructure_media`	`media_id` (PK), `ivfflat (embedding vector_cosine_ops)`	`facility_type`, `file_path`, `phash`, `dhash`, `ahash`, `whash` (64-bit hex), `embedding vector(128)`, `exif_json`, `geo_distance_m`, `geo_status`, `cv_detections_json`, `aph_compliance_json`, `clone_matches_json`.
`audit_logs`	`id` (Serial PK), Index on `app_id`	`timestamp`, `actor_role`, `actor_name`, `action_type`, `target_id`, `rule_id`, `ai_confidence`, `status_after`, `rationale`, `prev_hash` (SHA-256), `entry_hash` (SHA-256).

2. FastAPI REST Endpoint Specification (`backend/main.py`)

Method	Endpoint Route	Request Parameters & Response Payload
`GET`	`/api/health`	Returns engine readiness and active library versions (PyMuPDF, pikepdf, OpenCV, imagehash, exifread).
`GET`	`/api/applications`	Returns sorted list of institutional applications with risk scores, intake comparisons, and summary statuses.
`GET`	`/api/applications/{app_id}`	Returns complete dossier: institution profile, 4-way consistency table, APH checklist, live X-Ray document forensics, CV media analysis, and SHA-256 audit history.
`GET`	`/api/documents/{doc_id}/xray`	Query params: `quality` (40–98) & `scale` (5–45). Recomputes live OpenCV ELA heatmap, 8×8 noise grid, SIFT copy-move, and QR decode on demand.
`POST`	`/api/authority/verify/{doc_id}`	Triggers live verification sync with the simulated Issuing Authority portal and appends a signed SHA-256 audit log entry.
`POST`	`/api/scrutiny/decision`	Body: `app_id`, `target_id`, `target_type`, `new_status`, `rationale`. Updates status and commits a new SHA-256 block to the audit chain.
`POST`	`/api/live-analyze`	Multipart upload (any custom PDF/JPEG/PNG or `preset_specimen`). Executes full 3-Layer X-Ray Forensics + Clone Hunter + EXIF GPS verification in real time.

3. Engineered Synthetic Benchmark Dossiers (`backend/seed_generator.py`)

To enable rigorous, reproducible evaluation without exposing confidential institutional data, `seed_generator.py` programmatically builds real multi-feature PDFs and EXIF-tagged JPEGs on startup: (1) **`state_noc_8841.pdf`** (embedded QR with `approved_intake: 120`, Row 6 overwritten with `180 Seats` in `Courier` font over a quality-32 spliced JPEG patch, saved with Photoshop metadata and 2 `%%EOF` markers); (2) **`fire_safety_8841.pdf`** (QR `valid_upto: 2025-03-31` vs. printed `2026-12-31`, plus a duplicated Chief Fire Officer circular seal caught by SIFT); (3) **`floor_plan_8841.pdf`** (CAD vector blueprint where Classroom `CR-102` is `48.0 sq.m < 66.0 sq.m` norm); (4) **`affidavit_8841.pdf`** (clean e-Stamp with valid DSC); and (5) **`computer_lab_8841.jpg`** (cloned from Nagpur archive `deccan_historical_lab_3109.jpg` with Nagpur EXIF GPS coordinates).

What is Built (Current MVP) vs. What is To Be Implemented (Phase 2 Roadmap)

The table below provides a transparent, module-by-module breakdown of **what is 100% built and running in the current Veritas MVP** versus **what is planned for Phase 2 production deployment with AICTE**:

Module / Domain	Built & Working in Current MVP (v2.4)	To Be Implemented in Phase 2 (Production Scale)
1. OCR & Cross-Document Consistency	• Real `PyMuPDF` span-level extraction with bounding boxes. • 4-Way Consistency Matrix (App Form ↔ OCR ↔ QR/API ↔ AICTE DB). • APH 2024–27 configurable checklist rules (intake, area, FSR, TER).	• Integration of Vision-Language Models (**Qwen2.5-VL / Surya OCR**) for handwritten regional-language NOCs & degraded 100-DPI scans. • Multi-page table stitching across 100+ page trust deeds & land records.
2. 3-Layer Document Forensics	• Multi-pass OpenCV `QRCodeDetector` & QR-vs-OCR cross-lock tethers. • `pikepdf` metadata, `%%EOF` incremental save count & span font anomaly detector. • Real-time OpenCV JPEG ELA heatmap, 8×8 Laplacian Noise Grid & SIFT Copy-Move seal clone detector.	• Full cryptographic X.509 Certificate Revocation List (CRL / OCSP) & CCA root-chain validation via `pyHanko`. • Deep-learning splicing localization (**TruFor / CAT-Net** ViT weights) for non-JPEG PNG/TIFF scans. • C2PA cryptographic provenance verification.
3. Issuing Authority Verification	• Simulated connector endpoints for DigiLocker / State e-NOC, Municipal Fire Portal, COA Registry & StockHolding CRA. • All 5 statutory statuses + live re-verify ping & latency logging.	• Live production OAuth2 / mTLS integration with **API Setu**, **DigiLocker Requester API**, **Bharatkosh**, and State Fire Portals. • Automated webhook listeners for asynchronous "Pending → Verified" updates.
4. Infrastructure CV & Clone Hunter	• Real 64-bit `pHash`, `dHash`, `aHash`, `wHash` + 128-D feature vector search. • Real `exifread` WGS84 Haversine GPS distance verifier. • Side-by-side Cross-College Clone Comparator UI & APH room/equipment checks.	• Custom fine-tuned **YOLOv8x / RT-DETR** weights trained on 50,000+ Indian college classroom & lab images + **CLIP ViT-L/14** 768-D embeddings in distributed `pgvector`. • Video walkthrough frame sampling & monocular depth/room-area estimation.
5. Portal UI, Workflow & Audit Trail	• 5-Page Daylight Civic GovTech Portal (Queue, Checklist, Forensics Studio, Infrastructure CV, Upload & Audit). • Role Switcher (Scrutiny Officer vs. Nodal Officer). • Live Multipart Upload Sandbox & SHA-256 hash-chained audit ledger.	• SSO integration with **AICTE Web Portal / Parakh / MeriPehchaan (NSSO)**. • Distributed Celery + Redis worker queue for batch processing 10,000+ EoA dossiers. • Automated PDF Deficiency Show-Cause Notice generator with embedded forensic exhibits.

2. Prototype Evaluation Metrics & Benchmarks

Scrutiny Time Reduction

Reduces preliminary dossier scrutiny from **~40 minutes of manual cross-checking to < 45 seconds** of guided evidence review (~88% reduction in manual effort).

Forensic & Clone Recall

100% detection recall on QR-vs-OCR mismatches, incremental PDF updates, span font injections, and cross-college cloned photos (`D_H <= 12`).

Explainability & Governance

100% of AI flags carry a normalized bounding box, an APH 2024–27 clause reference, and a SHA-256 hash-chained officer override workflow.

Conclusion

Veritas delivers a complete, production-aligned solution for **SIH26261**—bridging multi-document OCR reconciliation, 3-layer document forensics, authoritative registry verification, and computer-vision infrastructure inspection inside an intuitive, daylight-first civic portal designed for AICTE officials.